

Multiple Choice

1. **Answer:** A

Options: A) Results in frequent urination and thirst; B) It is a result of an overactive thyroid gland; C) High cholesterol levels are a common symptom; D) It is a recognized cause of hypertension; E) It is caused by excessive production of insulin

Note: Addison's Disease leads to insufficient production of adrenal hormones, particularly cortisol, which disrupts the body's ability to regulate water and electrolyte balance, resulting in symptoms such as frequent urination and excessive thirst.

2. **Answer:** A

Options: A) Anencephaly; B) Normal pregnancy; C) Neural tube defects; D) Exomphalos; E) Cardiac defects

Note: Anencephaly, a severe neural tube defect, typically results in lower maternal serum levels of alpha-fetoprotein (AFP) due to the absence of brain tissue and skull, which normally produce AFP, leading to reduced levels detected in maternal circulation.

3. **Answer:** C

Options: A) If administered within 4 days of clinical signs; B) If used within 48 hours of first clinical signs; C) Used for the obese; D) Used in children under the age of 2 years where high virus spread is noted

Note: Antivirals can be used prophylactically or therapeutically in obese individuals due to their increased risk for severe infections and complications, necessitating preventive or therapeutic interventions to manage their health effectively.

4. **Answer:** A

Options: A) Fat conducts electricity better than muscle; B) Fat contains more electrolytes than muscle; C) Muscle weighs more than fat; D) Muscle is denser than fat; E) Fat is denser than muscle

Note: The correct answer is based on the fact that fat tissue has a lower water content and higher resistance to electrical current compared to muscle, which contains more water and electrolytes, allowing electricity to travel faster through muscle.

5. **Answer:** E

Options: A) Backward; B) Circular motion; C) No movement, it remains stationary; D) Toward the midline of the body; E) Downward

Note: The correct answer is "downward" because abduction of the foot refers to the movement away from the midline of the body, which in the context of the foot typically involves the toes moving away from the centerline, resulting in a downward direction.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the cardiac cycle includes a distinct phase of relaxation known as diastole, during which the heart chambers fill with blood, in addition to the contraction phase called systole.

7. **Answer:** False

Options: A) True; B) False

Note: HIV infection rates in Europe have not shown a consistent decrease due to various factors such as inequities in healthcare access, rising incidence in certain populations, and the ongoing stigma surrounding the virus, which hampers effective prevention and treatment efforts.

8. **Answer:** False

Options: A) True; B) False

Note: Coronaviruses are characterized by crown-like (corona) spike proteins that protrude from a lipid bilayer, not a rigid barrel-shaped structure, as they have an enveloped, spherical shape rather than a rigid form.

9. **Answer:** False

Options: A) True; B) False

Note: Noradrenaline is not released at the motor endplate; instead, acetylcholine is the neurotransmitter responsible for stimulating muscle contraction at this site.

10. **Answer:** False

Options: A) True; B) False

Note: The urethra primarily serves as a conduit for urine to exit the body, rather than a collection reservoir, which is the role of the bladder.

Long Form Response

11. **Answer:** Vitamin D plays a crucial role in enhancing calcium absorption from the small intestine, primarily through its active form, calcitriol. This hormone binds to vitamin D receptors in intestinal cells, promoting the synthesis of calcium transport proteins such as calbindin. Consequently, this increases the efficiency of calcium uptake, ensuring adequate levels for physiological functions, including bone health and muscle function. Insufficient vitamin D can lead to impaired calcium absorption, resulting in conditions such as osteomalacia or osteoporosis, which highlight the importance of maintaining adequate vitamin D levels for overall health. Therefore, understanding the mechanisms of vitamin D in calcium metabolism is essential for preventing related deficiencies and associated health complications.

Note: correct because it accurately describes the mechanism by which vitamin D,

specifically its active form calcitriol, enhances calcium absorption in the small intestine through the stimulation of calcium transport proteins, while also addressing the health implications of sufficient vitamin D levels and the consequences of deficiency.

12. **Answer:** The anatomical significance of locating the carotid pulse lies in its proximity to the carotid artery, a major vessel supplying blood to the brain, thereby serving as a critical indicator of cardiovascular health. Clinically, the carotid pulse is vital for assessing the patient's hemodynamic status, as it provides information on heart rate, rhythm, and the strength of cardiac output. In emergency situations, palpating the carotid pulse is essential for determining the presence of circulation, guiding interventions such as cardiopulmonary resuscitation (CPR) or the administration of medications. Understanding the implications of carotid pulse assessment can significantly influence patient outcomes, particularly in critical care settings where timely identification of cardiac compromise is crucial.

Note: correct because it emphasizes the carotid pulse's role as a crucial indicator of cardiovascular health and hemodynamic status, which is essential for health assessments and determining circulation in emergency situations.